

HybridSparse: An End-to-End Hybrid Framework for Efficient Large-Scale Retrieval

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Abstract

Large-scale retrieval systems must operate under strict latency constraints while maintaining high recall. Sparse retrieval offers efficiency and interpretability, whereas dense retrieval provides stronger semantic matching. Although hybrid approaches combine both signals, their interaction is often limited, especially under intersection-based retrieval. We introduce HybridSparse, an end-to-end hybrid retrieval framework that strengthens sparse-dense interaction across modeling, training, and serving. It adopts a unified encoder with a shared backbone and jointly optimizes lexical and semantic representations through co-training. To further improve alignment, we incorporate hybrid score regularization and consistency distillation, enabling more stable and effective hybrid scoring. Experiments on public benchmarks demonstrate consistent improvements over strong sparse, dense, and hybrid baselines. In large-scale production deployment for Bing advertisement retrieval, HybridSparse delivers a +1.30% RPM gain, highlighting its practical impact.

CCS Concepts

• **Information systems** → *Retrieval efficiency*; **Language models**; *World Wide Web*; • **Computing methodologies** → *Natural language processing*.

Keywords

Retrieval System; Hybrid Retrieval; Neural Encoder-based Retrieval

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1 Introduction

Large-scale retrieval under strict latency constraints is central to web search, dialogue, and recommendation [18, 24, 27]. Two paradigms dominate first-stage retrieval: *sparse* lexical matching, which is efficient and interpretable but suffers from vocabulary mismatch and limited semantics [8, 15], and *dense* semantic retrieval, which improves semantic matching via low-dimensional embeddings but may lose salient lexical signals due to compression [12]. With transformer-based PLMs [5, 16], dense retrieval has become practical with ANN indices [3, 11, 13], while sparse retrieval has been strengthened by learned term weighting and expansion [6, 7, 20]. These advances make hybrid retrieval appealing, but also expose a key bottleneck: hybrid effectiveness depends not only on having both signals, but on how well they interact during candidate generation and scoring.

Existing hybrid methods typically follow two lines. A common approach performs *separate* sparse and dense searches over different indices and then fuses or reranks results [14, 22, 23]. While straightforward, this introduces redundant computation, additional latency, and score misalignment across paradigms. Another line improves one paradigm using signals from the other (e.g., lexical-aware dense training or partial dense augmentation for sparse) [1, 10, 17, 25]. These approaches enhance representations, but the interaction is often limited by a dominant paradigm (e.g., hybrid used mainly at reranking), leaving the first-stage hybrid candidate overlap and score consistency under-optimized.

OneSparse [4] takes a major step towards efficient hybrid serving by unifying sparse terms and dense virtual terms in a single inverted index, enabling intersection-based candidate selection without running two independent retrieval pipelines. This design has proven effective at scale (e.g., in Bing Sponsored Search), but it also reveals an important limitation: structural unification does not guarantee *behavioral* alignment. In intersection-based hybrid retrieval, sparse and dense components still behave as largely independent scorers; when their retrieved candidates diverge, the intersection amplifies the misalignment and can cap recall.

To address this gap, we propose **HybridSparse**, an end-to-end hybrid framework that strengthens sparse–dense interaction throughout modeling, training, and search execution while retaining the efficiency of unified serving. HybridSparse uses a shared encoder backbone with two lightweight branches to produce sparse lexical weights and dense embeddings, enabling parameter sharing and consistent representations. We further introduce co-training to jointly optimize both branches, and incorporate hybrid score regularization and consistency distillation to explicitly align scoring signals, improving robustness under intersection-based candidate selection. Our retrieval pipeline builds on OneSparse’s unified index and multi-way merge, ensuring scalable deployment.

We evaluate HybridSparse on public benchmarks and large-scale production data, and deploy it in Bing advertisement retrieval, yielding a **+1.30%** RPM (Revenue Per Mille) improvement.

2 HybridSparse

This section presents our unified neural encoder, co-training framework, and hybrid-guided optimization to enhance sparse–dense synergy.

2.1 Neural Encoder

Our retrieval system adopts a dual-encoder architecture. We employ a unified encoder to jointly support dense and sparse retrieval. Dense and sparse embeddings are denoted as $\mathbf{q}_{\text{sem}}$, $\mathbf{q}_{\text{lex}}$ and $\mathbf{d}_{\text{sem}}$, $\mathbf{d}_{\text{lex}}$. As query and document encoders share the same structure, we describe the query side.

The encoder consists of a shared global backbone and two task-specific encoders. A tokenized query is first encoded into a shared hidden state $\mathbf{H}_{\text{glo}}$, which is then processed by the semantic and lexical encoders.

2.1.1 Global Backbone. Given a query

$$q = [\text{CLS}], q_0, \dots, q_n, [\text{SEP}], \quad (1)$$

the global backbone $\mathcal{F}_{\text{global}}$ produces

$$\mathbf{H}_{\text{glo}} = \mathcal{F}_{\text{global}}(q). \quad (2)$$

$\mathbf{H}_{\text{glo}}$ is shared by both encoders, enabling dense and sparse representations to be derived from a consistent feature space while reducing redundant computation.

2.1.2 Semantic Encoder. The semantic encoder $\mathcal{F}_{\text{sem}}$ refines $\mathbf{H}_{\text{glo}}$ and produces a dense embedding via

$$\mathbf{q}_{\text{sem}} = \text{DensePooler}(\mathcal{F}_{\text{sem}}(\mathbf{H}_{\text{glo}})), \quad (3)$$

where the [CLS] token is used as the final representation.

2.1.3 Lexical Encoder. The lexical encoder maps $\mathbf{H}_{\text{glo}}$ into a vocabulary-sized sparse embedding. Following SPLADE-style designs [6], token representations are projected to the vocabulary space and aggregated via sparse pooling:

$$\mathbf{q}_{\text{lex}} = \log(1 + \text{Max}(\text{ReLU}(\mathbf{W}\mathcal{F}_{\text{lex}}(\mathbf{H}_{\text{glo}})))). \quad (4)$$

FLOPS regularization [21] is applied to encourage sparsity.

2.1.4 Discussions. A unified encoder with a shared backbone ensures that sparse and dense representations are learned from a consistent feature space, reducing misalignment while maintaining efficiency and interpretability.

2.2 Optimization in HybridSparse

2.2.1 General objectives. Training data are organized as triplets (q, d^+, D_q^-) . We adopt a standard contrastive learning objective and define a softmax-normalized probability over the candidate set $D_q = \{d^+\} \cup D_q^-$:

$$P_*(d^+ | D_q) = \frac{\exp(s_*(q, d^+))}{\sum_{d \in D_q} \exp(s_*(q, d))}, \quad (5)$$

For semantic and lexical encoders, we minimize negative log-likelihood losses:

$$\mathcal{L}_{\text{sem}} = -\log P_{\text{sem}}(d^+ | D_q), \quad (6)$$

$$\mathcal{L}_{\text{lex}} = -\log P_{\text{lex}}(d^+ | D_q), \quad (7)$$

with FLOPS-based sparsity regularization to ensure sufficient sparsity in $\mathbf{q}_{\text{lex}}$ and $\mathbf{d}_{\text{lex}}$, where $l_{\text{flops}}(q) = \sum_{t \in V} \left(\frac{1}{B} \sum_{i=1}^B w_t^q \right)^2$ and similarly for d :

$$\mathcal{L}_{\text{flops}} = \lambda_q l_{\text{flops}}(q) + \lambda_d l_{\text{flops}}(d). \quad (8)$$

The co-training objective is

$$\mathcal{L}_{\text{co}} = \lambda_{\text{sem}} \mathcal{L}_{\text{sem}} + \lambda_{\text{lex}} \mathcal{L}_{\text{lex}} + \mathcal{L}_{\text{flops}}. \quad (9)$$

2.2.2 Hybrid Optimization. The final hybrid score is computed as

$$s_{\text{hy}} = w_{\text{sem}} s_{\text{sem}} + w_{\text{lex}} s_{\text{lex}}. \quad (10)$$

Hybrid Score Regularization. We treat s_{hy} as a retrieval score and compute its negative log-likelihood as a regularization term:

$$\mathcal{L}_{\text{reg}} = -\log P_{\text{hy}}(d^+ | D_q). \quad (11)$$

Consistency Distillation. The hybrid score further serves as a teacher signal, aligning semantic and lexical distributions via KL divergence:

$$\mathcal{L}_{\text{dis}} = \text{KL}(P_{\text{hy}}, P_{\text{sem}}) + \text{KL}(P_{\text{hy}}, P_{\text{lex}}). \quad (12)$$

The overall objective is

$$\mathcal{L}_{\text{hy}} = \mathcal{L}_{\text{co}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_{\text{dis}} \mathcal{L}_{\text{dis}}. \quad (13)$$

2.2.3 Discussions. Intersection-based hybrid retrieval critically depends on sparse–dense alignment to ensure sufficient candidate overlap. Co-training improves representation agreement, while hybrid score regularization and consistency distillation directly align scoring signals, leading to more stable and effective hybrid retrieval.

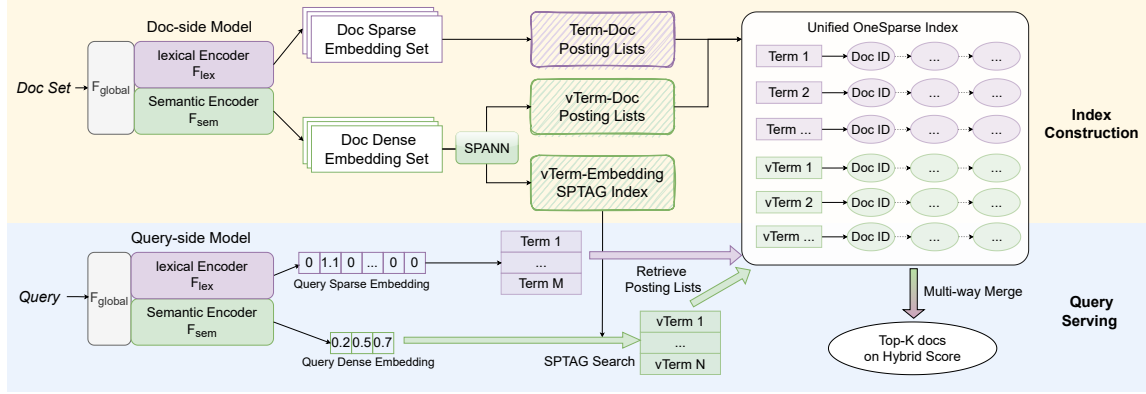


Figure 1: Illustration of the workflow of HybridSparse. A unified OneSparse Index is pre-built on the document set. During online retrieval, the query model generates dense and sparse embeddings, identifies relevant terms, retrieves virtual terms from the SPTAG index, and performs a multi-way merge on the OneSparse Index to obtain the final results.

2.3 Online Retrieval via HybridSparse

Leveraging the end-to-end hybrid retriever HybridSparse, we have architected a unified serving framework that employs a consolidated index system to manage both sparse and dense processes in hybrid search. As illustrated in Figure 1, the serving pipeline encompasses three stages, commencing with a user query.

- (1) Query dense embedding q_{sem} and sparse embedding q_{lex} are generated concurrently by the semantic and lexical encoders. q_{lex} is mapped to M terms to minimize latency.
- (2) q_{sem} is used to retrieve virtual terms via ANN search. The ANN index is partitioned across multiple machines at billion-scale. Retrieved cluster centroids are ranked, and only Top- N virtual terms are forwarded to reduce bandwidth.
- (3) Posting lists for both lexical terms and virtual terms are fetched from the inverted index. Final candidates are obtained through posting-list intersection and ranked using the hybrid score in Eq. 10. Similar to ANN search, the inverted index is document-partitioned across machines.

3 Public Dataset Evaluation

3.1 Dataset and Metrics

We evaluate on MSMARCO Passage Retrieval [19] and MSMARCO Web Search [2]. MSMARCO contains 8.8M passages with 7K dev queries; MS Web Search includes 101M documents and 9K test queries, reflecting large-scale multilingual search. We report MRR@10 and Recall. Baselines include BM25 and SPLADE (sparse), DPR, ANCE, and SimANS (dense), as well as OneSparse with a coCondenser-based dense encoder.

3.2 Implementation Detail

We adopt a PLM-based encoder with a shared backbone and two task-specific branches, initialized from coCondenser [9] on MSMARCO and SimANS [28] on MS Web Search. A pre-trained MLM head supports sparse term weighting and co-training initialization. Hybrid scoring uses direct score summation without weight tuning due to hybrid score regularization.

3.3 Experimental Results

3.3.1 Main Results. Table 1 shows that HybridSparse consistently outperforms all sparse and dense baselines in end-to-end hybrid retrieval. Compared to OneSparse, it achieves +1.7% and +2.3% Recall gains on MSMARCO and MS Web Search, together with +2.0% and +2.5% MRR improvements.

Beyond overall hybrid performance, HybridSparse also improves the individual sparse and dense components. PLM-based lexical modeling substantially enhances recall over BM25, while dense embeddings trained under our joint framework outperform OneSparse’s dense encoder, demonstrating the benefit of unified optimization.

4 Offline Experiments on Production Data

4.1 Dataset and Metrics

Dataset. A large-scale dataset is curated from Bing’s production for offline experiments. This dataset comprises a total of 1 billion query-ad pairs sourced from real-world users, with 1 million pairs reserved for testing purposes. Additionally, 30 million ads are incorporated into the ANN index for embedding based retrieval (EBR) evaluation.

Evaluation Metrics. The following two metrics are employed to evaluate retrieval quality: 1) Recall of click (R@K): This metric assesses whether the clicked advertisement is included within the Top-K retrieval results, analogous to recall in retrieval tasks, where the ground truth is defined by the click label. 2) Relevance degree (Rel@K): This metric evaluates the average relevance score for the Top-K retrieval results. The relevance labels are predicted by a high-precision LLM-based relevance teacher, which is calibrated on human-annotated query-ad pairs and evaluated against human judgments before being used for offline evaluation. To reduce evaluation variance, we use a fixed scoring schema and deterministic inference settings.

4.2 Implementation Detail.

The dense embedding’s dimensionality is reduced from 768 to 64 through a linear projection. We utilize a WordPiece tokenizer, trained on an in-house ads corpus to optimize performance, with

Table 1: Overall results on MSMARCO and MS Web Search. Subscripts denote standalone results; [S]/[D] indicate sparse/dense baselines. Dense uses SPANN index.

Method	MSMARCO				MSMARCO Web Search			
	MRR@10↑	Recall@1↑	Recall@10↑	Recall@100↑	MRR@10↑	Recall@1↑	Recall@10↑	Recall@100↑
BM25 ^[S]	0.1928	0.1267	0.3747	0.6502	0.2960	0.2230	0.4600	0.6387
SPLADE ^[S]	0.3422	0.1969	0.6159	0.8759	0.5679	0.5058	0.7511	0.8393
ANCE ^[D]	0.3297	0.1999	0.5710	0.8329	0.5800	0.4987	0.7294	0.8018
SimANS ^[D]	0.3870	0.2530	0.6370	0.8567	0.5850	0.5063	0.7314	0.7982
OneSparse	0.3800	0.2544	0.6683	0.8958	0.6142	0.5322	0.7660	0.8403
OneSparse _{sparse}	0.1928	0.1267	0.3747	0.6492	0.3890	0.3044	0.5677	0.7235
OneSparse _{dense}	0.3766	0.2371	0.6354	0.8766	0.5895	0.5068	0.7325	0.8013
HybridSparse	0.3877	0.2618	0.6750	0.9081	0.6278	0.5443	0.7796	0.8524
HybridSparse _{sparse}	0.3735	0.2308	0.6414	0.8825	0.5793	0.5192	0.7627	0.8500
HybridSparse _{dense}	0.3777	0.2421	0.6380	0.8777	0.5952	0.5168	0.7400	0.8092

Table 2: Offline testing results on production dataset.

Method	Rel@50↑	R@50↑
OneSparse _{BM25}	0%	0%
OneSparse _{SPLADE}	+16.45%	+16.56%
HybridSparse	+38.33%	+26.05%
HybridSparse _{dense}	+11.61%	+8.96%
HybridSparse _{sparse}	+21.91%	+11.34%

Table 3: Ablation study on production dataset

Method	Rel@50↑	R@50↑
OneSparse _{BM25}	0%	0%
HybridSparse	+38.33%	+26.05%
w/o Two-stage Fine-tuning	+37.08%	+21.76%
w/o Distillation	+37.12%	+26.47%

a vocabulary size of 50,777. In real-world scenarios, beyond optimizing recall, ensuring the relevance of retrieved content is crucial. Therefore, we employed a two-stage training process to optimize the model, where the second stage leverages a relevance teacher to self-generate and select hard negative samples, following [26]. For online serving, the dense and sparse query representations are produced in a single forward pass and consumed by the existing partitioned retrieval infrastructure. The two retrieval branches are executed concurrently, and the final merge follows the production OneSparse serving pipeline. Thus, the proposed hybrid optimization mainly changes model training and representation alignment, without introducing an additional online model stage. The models are implemented using Python 3.8 and PyTorch 1.12.1, and are trained on machines equipped with 8 NVIDIA V100 32GB GPUs and 4 AMD EPYC 7V12 64-Core CPUs.

4.3 Experimental Results Analysis

4.3.1 Overall Analysis. As shown in Table 2, OneSparse_{BM25}, which represents the vanilla OneSparse, serves as the baseline, having been previously deployed in production. This model employs BM25 for the sparse index and a semantic encoder without hybrid training for the dense index, resulting in the lowest performance. An enhancement to OneSparse involves replacing BM25 with SPLADE, referred to as OneSparse_{SPLADE}. This model is trained using query-ad pairs, leading to substantial performance improvements. Our proposed HybridSparse incorporates hybrid training for both the dense

and sparse sides, along with regularization and distillation to align the distributions of the two sides. This approach significantly enhances Recall and Relevance metrics compared to OneSparse_{SPLADE}, demonstrating the effectiveness of our hybrid training method. Additionally, HybridSparse_{dense} and HybridSparse_{sparse}, which represent the individual components extracted from HybridSparse, show considerable improvements over OneSparse_{BM25}. This indicates that hybrid training also provides individual benefits for each component of our hybrid method.

4.3.2 Ablation Study. As shown in Table 3, we conducted an ablation study to investigate the impacts of removing two-stage fine-tuning and distillation. The results indicate that without two-stage fine-tuning, both Recall and Relevance metrics experience a decline. The larger loss in Relevance may be attributed to the absence of relevance negative samples. Furthermore, removing distillation slightly improves Rel@50 but hurts R@50, revealing a trade-off between click-oriented recall and relevance. Since missed candidates cannot be recovered by downstream ranking stages, we prioritize the distillation setting that improves R@50 while keeping the relevance drop small. This suggests that the distillation weight can be tuned to balance recall and relevance under different production requirements.

Table 4: Revenue Per Mille (RPM) growth in A/B test, with each version’s RPM representing the additional gain over the previous version.

Method	RPM Growth↑	P-Value
OneSparse _{SPLADE}	1.29%	7.47×10^{-3}
HybridSparse	1.30%	1.72×10^{-3}

5 Online A/B Tests

The online impact of HybridSparse is presented in Table 4. Before deploying HybridSparse, an intermediate version, OneSparse_{SPLADE}, was implemented in Bing’s production environment. This version, which utilized query-ad pairs trained with SPLADE for OneSparse, achieved a 1.29% RPM growth in A/B testing, with a P-Value of 7.47×10^{-3} (a P-Value below 5×10^{-2} is considered significant in production). Ultimately, with the incorporation of hybrid training and self-distillation in our proposed HybridSparse, the HybridSparse

version resulted in an additional 1.30% RPM gain, building upon the improvements of the OneSparse_{SPLADE}.

6 Conclusion

We introduce HybridSparse, an end-to-end hybrid retrieval framework with unified neural encoding and joint optimization to strengthen sparse–dense interaction. A shared encoder enables efficient joint modeling, while hybrid score regularization and consistency distillation improve alignment and stability. Experiments on public and production data show consistent gains over strong baselines. In Bing advertisement retrieval, HybridSparse achieves a +1.30% RPM improvement at production scale.

References

- [1] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. *arXiv:2402.03216 [cs.CL]* <https://arxiv.org/abs/2402.03216>
- [2] Qi Chen, Xiubo Geng, Corby Rosset, Carolyn Buracton, Jingwen Lu, Tao Shen, Kun Zhou, Chenyan Xiong, Yeyun Gong, Paul Bennett, et al. 2024. MS MARCO Web Search: a Large-scale Information-rich Web Dataset with Millions of Real Click Labels. In *Companion Proceedings of the ACM on Web Conference 2024*. 292–301.
- [3] Qi Chen, Bing Zhao, Haidong Wang, Mingqin Li, Chuanjie Liu, Zengzhong Li, Mao Yang, and Jingdong Wang. 2021. Spann: Highly-efficient billion-scale approximate nearest neighborhood search. *Advances in Neural Information Processing Systems* 34 (2021), 5199–5212.
- [4] Yaoqi Chen, Ruicheng Zheng, Qi Chen, Shuotao Xu, Qianxi Zhang, Xue Wu, Weihao Han, Hua Yuan, Mingqin Li, Yujing Wang, et al. 2024. OneSparse: A Unified System for Multi-index Vector Search. In *Companion Proceedings of the ACM on Web Conference 2024*. 393–402.
- [5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [6] Thibault Formal, Carlos Lassance, Benjamin Piwowarski, and Stéphane Clinchant. 2021. SPLADE v2: Sparse lexical and expansion model for information retrieval. *arXiv preprint arXiv:2109.10086* (2021).
- [7] Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant. 2021. SPLADE: Sparse lexical and expansion model for first stage ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2288–2292.
- [8] George W. Furnas, Thomas K. Landauer, Louis M. Gomez, and Susan T. Dumais. 1987. The vocabulary problem in human-system communication. *Commun. ACM* 30, 11 (1987), 964–971.
- [9] Luyu Gao and Jamie Callan. 2021. Unsupervised corpus aware language model pre-training for dense passage retrieval. *arXiv preprint arXiv:2108.05540* (2021).
- [10] Luyu Gao, Zhuyun Dai, Tongfei Chen, Zhen Fan, Benjamin Van Durme, and Jamie Callan. 2021. Complement lexical retrieval model with semantic residual embeddings. In *Advances in Information Retrieval: 43rd European Conference on IR Research, ECIR 2021, Virtual Event, March 28–April 1, 2021, Proceedings, Part I* 43. Springer, 146–160.
- [11] Aristides Gionis, Piotr Indyk, Rajeev Motwani, et al. 1999. Similarity search in high dimensions via hashing. In *Vldb*, Vol. 99. 518–529.
- [12] Jiafeng Guo, Yinqiong Cai, Yixing Fan, Fei Sun, Ruqing Zhang, and Xueqi Cheng. 2022. Semantic models for the first-stage retrieval: A comprehensive review. *ACM Transactions on Information Systems (TOIS)* 40, 4 (2022), 1–42.
- [13] Suhas Jayaram Subramanya, Fnu Devvrit, Harsha Vardhan Simhadri, Ravishankar Krishnawamy, and Rohan Kadekodi. 2019. Diskann: Fast accurate billion-point nearest neighbor search on a single node. *Advances in Neural Information Processing Systems* 32 (2019).
- [14] Saar Kuzi, Mingyang Zhang, Cheng Li, Michael Bendersky, and Marc Najork. 2020. Leveraging semantic and lexical matching to improve the recall of document retrieval systems: A hybrid approach. *arXiv preprint arXiv:2010.01195* (2020).
- [15] Hang Li, Jun Xu, et al. 2014. Semantic matching in search. *Foundations and Trends® in Information Retrieval* 7, 5 (2014), 343–469.
- [16] Yinhan Liu, Mylène Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692* (2019).
- [17] Yi Luan, Jacob Eisenstein, Kristina Toutanova, and Michael Collins. 2021. Sparse, dense, and attentional representations for text retrieval. *Transactions of the Association for Computational Linguistics* 9 (2021), 329–345.
- [18] Bhaskar Mitra, Fernando Diaz, and Nick Craswell. 2017. Learning to match using local and distributed representations of text for web search. In *Proceedings of the 26th international conference on world wide web*. 1291–1299.
- [19] Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. 2017. MS MARCO: A Human-Generated Machine Reading Comprehension Dataset. <https://openreview.net/forum?id=Hk1IOLcle>
- [20] Rodrigo Nogueira, Jimmy Lin, and AI Epistemic. 2019. From doc2query to docTTTTTquery. *Online preprint* 6, 2 (2019).
- [21] Biswajit Paria, Chih-Kuan Yeh, Ian EH Yen, Ning Xu, Pradeep Ravikumar, and Barnabás Póczos. 2020. Minimizing flops to learn efficient sparse representations. *arXiv preprint arXiv:2004.05665* (2020).
- [22] Minjoon Seo, Jinhyuk Lee, Tom Kwiatkowski, Ankur P Parikh, Ali Farhadi, and Hannaneh Hajishirzi. 2019. Real-time open-domain question answering with dense-sparse phrase index. *arXiv preprint arXiv:1906.05807* (2019).
- [23] Jianguo Wang, Xiaomeng Yi, Rentong Guo, Hai Jin, Peng Xu, Shengjun Li, Xianguyu Wang, Xiangzhou Guo, Chengming Li, Xiaohai Xu, et al. 2021. Milvus: A purpose-built vector data management system. In *Proceedings of the 2021 International Conference on Management of Data*. 2614–2627.
- [24] Jason Weston, Emily Dinan, and Alexander H Miller. 2018. Retrieve and refine: Improved sequence generation models for dialogue. *arXiv preprint arXiv:1808.04776* (2018).
- [25] Yingrui Yang, Parker Carlson, Shanxiu He, Yifan Qiao, and Tao Yang. 2024. Cluster-based partial dense retrieval fused with sparse text retrieval. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2327–2331.
- [26] Jianjin Zhang, Zheng Liu, Weihao Han, Shitao Xiao, Ruicheng Zheng, Yingxia Shao, Hao Sun, Hanqing Zhu, Premkumar Srinivasan, Weiwei Deng, et al. 2022. Uni-retriever: Towards learning the unified embedding based retriever in bing sponsored search. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 4493–4501.
- [27] Yuan Zhang, Xue Dong, Weijie Ding, Biao Li, Peng Jiang, and Kun Gai. 2023. Divide and conquer: Towards better embedding-based retrieval for recommender systems from a multi-task perspective. In *Companion Proceedings of the ACM Web Conference 2023*. 366–370.
- [28] Kun Zhou, Yeyun Gong, Xiao Liu, Wayne Xin Zhao, Yelong Shen, Anlei Dong, Jingwen Lu, Rangan Majumder, Ji-Rong Wen, Nan Duan, et al. 2022. Simans: Simple ambiguous negatives sampling for dense text retrieval. *arXiv preprint arXiv:2210.11773* (2022).